ABHIJEET KISHORE NAYAK

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EDUCATION

- M.Sc. - Computer Science, The University of Freiburg (**Grade: 1.4**) *04/2021 – 09/2023*
  - **Courses:** Computer Vision, Mobile Robotics, Pattern Recognition, Deep Learning, Numerical Optimization, Numerical Optimal Control
  - **Master Project:** Compositional Servoing by Recombining Demonstrations (Paper, Website)
    - Performing **robot manipulation** tasks with a minimum number of demonstrations
    - Enable skill sharing between different tasks by defining **demonstration trajectories**
  - **Master Thesis:** Radar Localization on Prior Lidar Maps (Paper, Website, Video)
    - **Cross-modal localization** using radar sensors for perception and lidar maps of the environment
    - Coarse-to-fine strategy – Global localization (without GPS) followed by metric localization
- B.E – Electronics and Communication, R. V. College of Engineering **(Grade: 8.97/10.0)** *08/2013 – 08/2017*
  - **Courses:** Applied Mathematics, Digital Signal Processing, Digital Circuits
  - **Thesis:** Generation of Simulated Inputs to Image-based Advanced Driver Assistance Systems

PROFESSIONAL EXPERIENCE

- Student Assistant – Production Control Group, Fraunhofer IPM *09/2022 – 10/2023*
  - Software development for the Freefall Inspection System
  - Machine Learning applications in the detection of anomalies in industrial parts
  - Configured a CIFX device with Profinet IO for real-time communication
  - Developed QT applications for better data visualization that helped improve product efficiency
- Teaching Assistant – Computer Graphics and Image Processing Course *04/2022 – 08/2022*
  - Designed and conducted practical exercise sessions for a class of 50+ students
- Research Intern – National Institute of Informatics, Japan *12/2021 – 04/2022*
  - NII International Internship Program, 2021
  - **3rd rank** in the Drone-vs-Bird Detection Challenge
- Engineer – Mercedes-Benz Research & Development India *09/2017 – 04/2021*
  - Developed algorithms for **lane localization** of the vehicle for a *Level-3 Automated driving project*
  - Conceptualized and developed algorithms for **fusion and tracking of static objects**
  - Automated **large-scale analysis** of test-drive data along with the creation of a dashboard
  - **Vehicle set up** with various sensors to collect data for a mapping research project
  - Generated a map in real time using state-of-the-art **SLAM** algorithms
  - Trained networks such as ERFNET to detect lane markings on the test-drive data

SKILLS

- C / C++ / Python Programming
- Robot Operating System
- Apache Spark (PySpark)
- State Estimation
- OpenCV
- PyTorch / Tensorflow
- Geospatial Analysis
- SQL
- Data Analysis
- Deep Learning
- Algorithm Design
- Object Tracking / Fusion

PUBLICATIONS

- **Nayak, A.**, Cattaneo, D., & Valada, A. (2023). RaLF: Flow-based Global and Metric Radar Localization in LiDAR Maps. arXiv preprint arXiv:2309.09875 (submitted to ICRA 2024)
- Argus, M., **Nayak, A.**, Büchner, M., Galesso, S., Valada, A., & Brox, T. (2023). Compositional Servoing by Recombining Demonstrations. arXiv preprint arXiv:2310.04271 (submitted to ICRA 2024)
- **Nayak, A**, et al. Evaluation of Fully Convolutional One-Stage Object Detection for Drone Detection. International Conference on Image Analysis and Processing. Cham: Springer International Publishing, 2022
- L. Kenye, R. Palugulla, M. Arora, B. Bhat, R. Kala, and **A. Nayak**, Re-localization for Self-Driving Cars using Semantic Maps, 2020 Fourth IEEE International Conference on Robotic Computing (IRC)
- Patent: A System and Method for Lane-Determination of a Vehicle (published)
  - Using **sensor fusion** to define trajectories of dynamic objects (E.g., Neighboring vehicles)
  - Utilize these defined trajectories and the HD map to localize the vehicle on the road
  - Implementation is functional on series production **Mercedes-Benz S-Class 2021**

HONOURS AND AWARDS

- 1st Place, Deep Learning Competition – Large Networks Track, University of Freiburg *03/2022*
- Inventor Trophy – Daimler AG *05/2019*
  - Appreciation for inventive achievements
- Bronze Star Award – Mercedes-Benz Research & Development India *12/2019*
  - Award for the best performer of the department